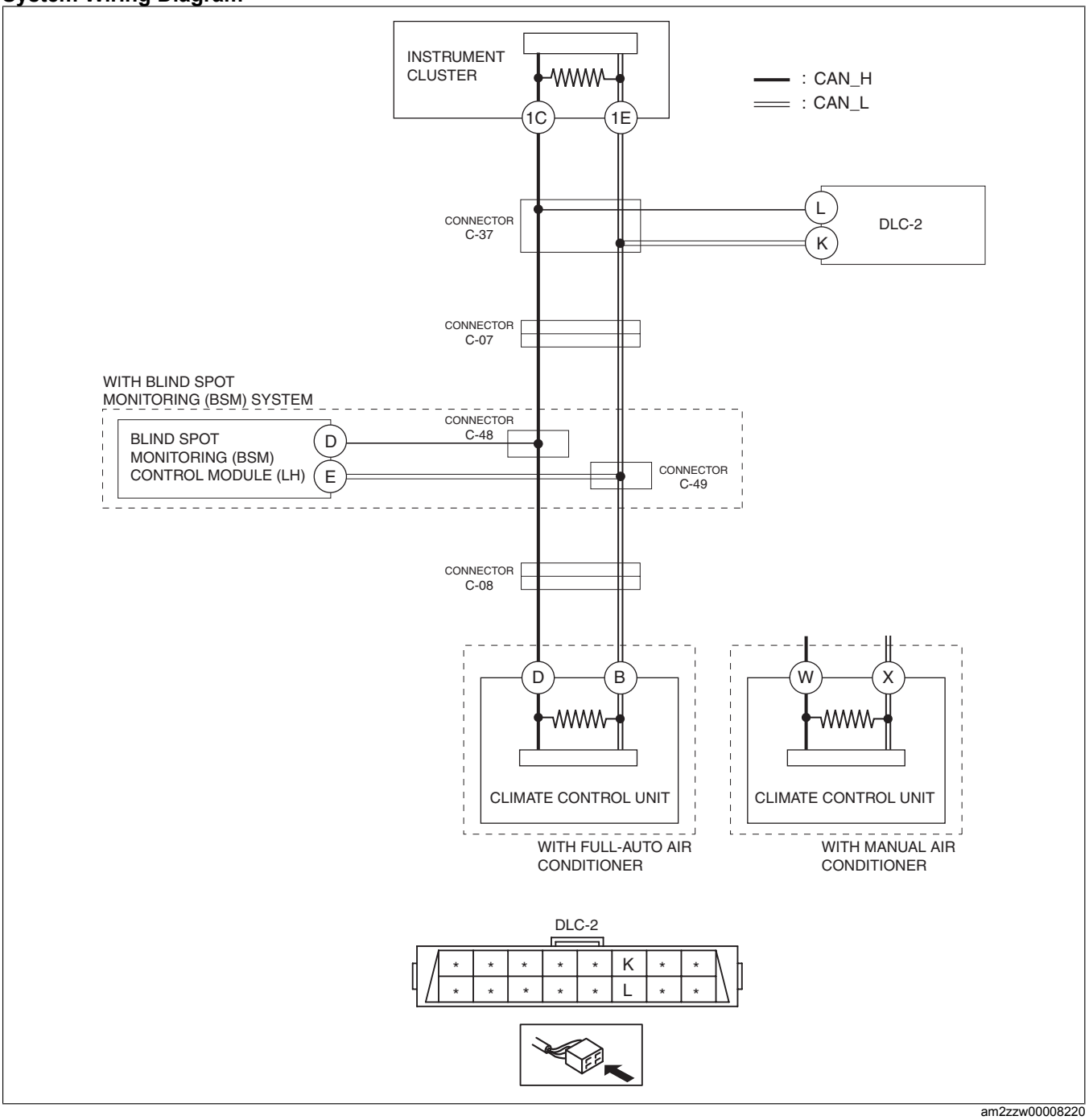
DETERMINING SHORT TO GROUND LOCATION (MS-CAN) [SKYACTIV-G 1.3, SKYACTIV-G 1.5 (R.H.D.)]

id100220000900

Caution

- **Perform the following malfunction diagnosis only when it is diagnosed with a short to ground by CONTROLLER AREA NETWORK (CAN) MALFUNCTION DIAGNOSIS FLOW. (See CONTROLLER AREA NETWORK (CAN) MALFUNCTION DIAGNOSIS FLOW [SKYACTIV-G 1.3, SKYACTIV-G 1.5 (R.H.D.)].)**

System Wiring Diagram



am2zzw00008220

Determination Procedure

Caution

- **When disconnecting the connector, verify that there is no looseness, damage, deformation, corrosion, or poor connection of the connector terminals.**

- **When inspecting the DLC-2, touch it with a paper clip or similar thin pin without directly inserting a tester into the terminals.**

Step	Inspection		Action
1	**INSPECT FOR SHORT TO GROUND BETWEEN CONNECTOR C-37 AND INSTRUMENT CLUSTER** • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the connector C-37. • Inspect for continuity at the following terminals: — Between instrument cluster terminal 1C and body ground — Between instrument cluster terminal 1E and body ground • Is there continuity?	Yes	Go to the next step.
		No	Go to Step 3.
2	**INSPECT CAN LINE IN INSTRUMENT CLUSTER FOR SHORT TO GROUND** • Disconnect the instrument cluster connector. • Inspect for continuity at the following terminals: — Between instrument cluster terminal 1C (wiring harness side) and body ground — Between instrument cluster terminal 1E (wiring harness side) and body ground • Is there continuity?	Yes	Repair or replace the wiring harness between the instrument cluster and connector C-37 because the wiring harness is shorted to ground.
		No	Replace the instrument cluster because there is a short to ground in the instrument cluster. (See INSTRUMENT CLUSTER REMOVAL/ INSTALLATION.)
3	**INSPECT FOR SHORT TO GROUND BETWEEN CONNECTORS C-37 AND DLC-2** • Inspect for continuity at the following terminals: — Between DLC-2 terminal L and body ground — Between DLC-2 terminal K and body ground • Is there continuity?	Yes	Repair or replace the wiring harness between connector C-37 and DLC-2 because the wiring harness is shorted to ground.
		No	Go to the next step.
4	**INSPECT FOR SHORT TO GROUND BETWEEN CONNECTOR C-37 AND CONNECTOR C-07** • Connect connector C-37. • Disconnect connector C-07. • Inspect for continuity at the following terminals: — Between DLC-2 terminal L and body ground — Between DLC-2 terminal K and body ground • Is there continuity?	Yes	Repair or replace the wiring harness between connector C-37 and connector C-07 because the wiring harness is shorted to ground.
		No	Go to the next step.
5	**INSPECT FOR SHORT TO GROUND BETWEEN CONNECTOR CONNECTORS C-48 AND C-49 AND CONNECTOR C-07** • Connect connector C-07. • Disconnect connectors C-48 and C-49. • Inspect for continuity at the following terminals: — Between DLC-2 terminal L and body ground — Between DLC-2 terminal K and body ground • Is there continuity?	Yes	Repair or replace the wiring harness between connector C-07 and connectors C-48 and C-49 because the wiring harness is shorted to ground.
		No	Go to the next step.

Step	Inspection		Action
6	**INSPECT FOR SHORT TO GROUND BETWEEN BLIND SPOT MONITORING (BSM) CONTROL MODULE (LH) AND CONNECTORS C-48 AND C-49** • Inspect for continuity at the following terminals: — Between blind spot monitoring (BSM) control module (LH) terminal D and body ground — Between blind spot monitoring (BSM) control module (LH) terminal E and body ground • Is there continuity?	Yes	Go to the next step.
		No	Go to Step 8.
7	**INSPECT CAN LINE IN BLIND SPOT MONITORING (BSM) CONTROL MODULE (LH) FOR SHORT TO GROUND** • Disconnect the blind spot monitoring (BSM) control module (LH) connector. • Inspect for continuity at the following terminals: — Between blind spot monitoring (BSM) control module (LH) terminal D (wiring harness side) and body ground — Between blind spot monitoring (BSM) control module (LH) terminal E (wiring harness side) and body ground • Is there continuity?	Yes	Repair or replace the wiring harness between the blind spot monitoring (BSM) control module (LH) and connectors C-48 AND C-49 because the wiring harness is shorted to ground.
		No	Replace the blind spot monitoring (BSM) control module (LH) because there is a short to ground in the blind spot monitoring (BSM) control module (LH). (See BLIND SPOT MONITORING (BSM) CONTROL MODULE REMOVAL/INSTALLATION.)
8	**INSPECT FOR SHORT TO GROUND BETWEEN CONNECTORS C-48 AND C-49 AND CONNECTOR C-08** • Connect connectors C-48 and C-49. • Connect connector C-08. • Inspect for continuity at the following terminals: — Between DLC-2 terminal L and body ground — Between DLC-2 terminal K and body ground • Is there continuity?	Yes	Repair or replace the wiring harness between connectors C-48 and C-49 and connector C-08 because the wiring harness is shorted to ground.
		No	Go to the next step.
9	**INSPECT CAN LINE IN CLIMATE CONTROL UNIT FOR SHORT TO GROUND** • Disconnect the climate control unit connector. • Inspect for continuity at the following terminals: — Between climate control unit terminal D (wiring harness side) and body ground (with full-auto air conditioner) — Between climate control unit terminal B (wiring harness side) and body ground (with full-auto air conditioner) — Between climate control unit terminal W (wiring harness side) and body ground (with manual air conditioner) — Between climate control unit terminal X (wiring harness side) and body ground (with manual air conditioner) • Is there continuity?	Yes	Repair or replace the wiring harness between the climate control unit and connector C-08 because the wiring harness is shorted to ground.
		No	Replace the climate control unit because there is a short to ground in the climate control unit. (See CLIMATE CONTROL UNIT REMOVAL/INSTALLATION [FULL-AUTO AIR CONDITIONER].) (See CLIMATE CONTROL UNIT REMOVAL/INSTALLATION [MANUAL AIR CONDITIONER].)